

SIAM MOSHARRAF HOSSAIN

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OBJECTIVE

Seeking an entry-level position in electrical engineering, to dedicate my experience and skills to thrive on it.

EDUCATION

University of Houston, Houston, TX

Dec 2011

Master of Science, Electrical Engineering, CGPA: 3.57

Islamic University of Technology, Dhaka, Bangladesh

Oct 2007

Bachelor of Science, Electrical & Electronic Engineering, CGPA: 3.78

Relevant Courses: Industrial Power System Protection, VLSI Design, IC Engineering, MEMS & Nano Devices, Solar Cells

EXPERIENCE

University of Houston, Houston, TX

Oct 2009 - Present

Facility Manager & Safety Designee, Electrochemical Nanofabrication and Nanomaterials Synthesis Group, ECE Dept.

- Responsible for training and smooth operation of a thin film stress lab and implementation of safety protocol
- Successfully designed and executed a closed loop experimental setup integrated with PID controller and LabVIEW, via multifunction data acquisition device, for a project by Texas Instruments
- Hands-on experience in performing annealing on electrodeposited magnetic thin film alloys (NiFe, CoNiFe, CoNiFeX) for in-situ stress-temperature analysis
- Experience in both independent and team research with application in variety of fields

Biman Bangladesh Airlines Ltd., Dhaka, Bangladesh

Apr 2008 - Jun 2009

Aircraft Maintenance Engineer, Avionics Dept.

- Hands-on experience in maintenance of avionic electrical system, electronic instrument system, aircraft power plant, communications and navigation system of DC-10, F-28 and Cessna aircrafts

SKILLS

- **Programming:** C, C++, Visual Basic, MATLAB, LabVIEW, Microchip MPLAB, MASM 8086, PIC16F84A, HTML
- **Simulation:** COMSOL, SKM, PSpice, DSCH, Simulink, ATHENA, ATLAS, PC1D, AppCAD, CALP
- **Software:** MySQL, MS Access, MS Visio, Origin, 3ds Max, Adobe Premiere Pro, Adobe Photoshop, Joomla, Wordpress
- **Material Characterization Tools:** SEM, AFM, VSM, Sputtering, XRD, XPS, Ellipsometer, Profilometer, Thermal Evaporator
- **Machine Tools:** Lathe Machine, Welding, Filing, Drilling, Drafting, Etching, Electrochemical Deposition

PROJECTS

- Examined a set of wavelet functions utilizing MATLAB and C/C++ for improving the level of compression on digital image
- Formulated a user-interactive graphical framework for Morphology evolution of Copper during Electropolishing using MATLAB
- Created layout and simulation of a 16-bit Serial input parallel output shift register using UNIX, CALP, Cadence and PSpice
- Created a model of an Arithmetic Logic Unit capable of doing complex mathematical and logical operation using DSCH and performed hardware implementation
- Designed a low-cost traffic signal system prototype and a digital clock with self error detection ability using PIC16F84A MCU, and designed a Bluetooth transceiver using AppCAD
- Designed a power plant using SKM following NEC, OSHA and IEEE standards, and produced documentation (calculations, sketches and data sheets) for equipment coordination, plant protection and control
- Analyzed steady-state and transient-state operation of a 3-phase power system with generator, synchronous motor & bus network
- Built a Micro-robot model based on arrays of movable silicon legs using COMSOL
- Developed several e-commerce websites using HTML, Joomla and Wordpress

PUBLICATIONS

- K.S. Martirosyan, E. Galstyan, **S. M. Hossain**, Yi-Ju Wang and D. Litvinov, "Barium Hexaferrite Nanoparticles: Synthesis and Magnetic Properties", *Materials Sc. and Engr. B*, 176 (2011), pp. 8–13
- **S. M. Hossain**, M. Y. Razib, I. Karim, "Improvement of Image Compression Using Wavelet Transform", *VDM Publishing* (2010)

AWARDS AND HONORS

- Competitive Graduate Assistant Tuition Waiver from University of Houston Jan 2010 – Dec 2011
- OIC Scholarship from Organization of Islamic Conference Jan 2003 – Oct 2007
- HSC excellence scholarship from national education board Jan 2003 – Oct 2007

ACTIVITIES AND INTERESTS

- IEEE
- Founder of HigherStudyAbroad™, a nonprofit academic advising network
- Professional freelance Video Editor & Photographer, edited 3 full-length movies, and promotionals